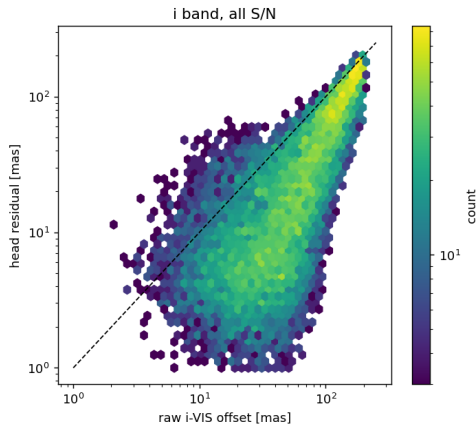
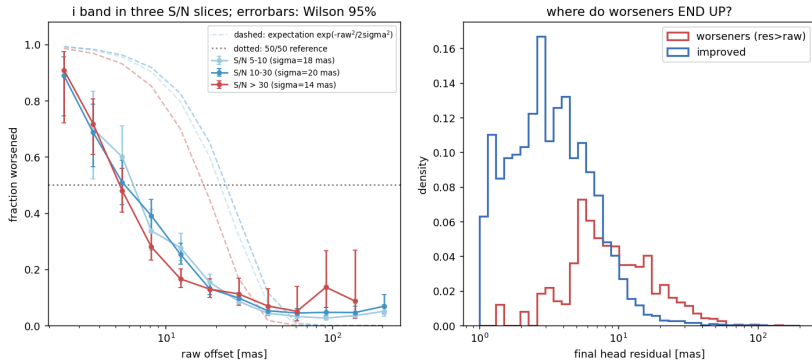


# Why Astrometry Is Degraded for a Subset of Objects: Diagnosis and Cure in JAISP

July 2026 (internal, ECDFS Q1 production head)



- The head improves the population (raw  $\sim 48$  mas  $\rightarrow$  13.4 mas median), but 9% of all measurements end up **farther** from VIS than they started.
- Two candidate explanations: (a) those sources were lucky before, and the small raw offset was chance; (b) the model fails on those sources specifically.



- Left: the same pattern in every S/N slice, each observed curve below its own re-measurement expectation (dashed,  $P(\text{worse}) = e^{-\text{raw}^2/2\sigma^2}$ ); pooled over bands, shared photons preserve lucky agreement more often than chance predicts.
- Only the **bright** slice grows a tail at large raw offsets, where chance predicts zero. Right: worseners sit right of the improved because the selection forces it (res > raw truncates the floor distribution from below); the truncated-floor prediction is 21.6 mas median versus 17.8 observed. Only the excess beyond  $\sim 40$  mas escapes any chance model.

For one source, per band  $b$ : fit the classical centroid, form  $\text{raw}_b = \text{pos}_b - \text{pos}_{\text{VIS}}$ ; the head predicts a correction  $c_b$  from one shared representation;  $\text{resid}_b = \text{raw}_b - c_b$ . “Strongly worsened in  $b$ ” =  $|\text{resid}_b| > 1.5 |\text{raw}_b|$  and worse by  $> 10$  mas. Rate: 5.6% of bright measurements.

- **Independent between bands:** the centroid-fit noise of each band image (different exposures, different photons). If this alone drove worsening, hits on the same source would be independent coin flips:  $P(3 \text{ hits}) \approx 0.056^3 \times 10 \text{ triples} \times 3185 \text{ sources} \approx 5$ .
- **Shared for one source:** (1) the VIS reference position, subtracted from every band; (2) the local WCS terms (Rubin bands share Rubin’s solution, NISP its own); (3) the source itself (proper motion, colour gradients); (4) the head’s shared representation, feeding all nine corrections.

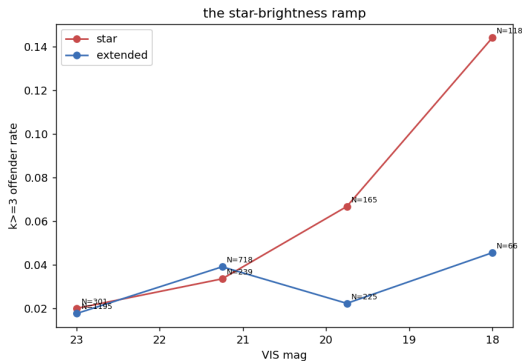
| strongly worsened in $\geq k$ Rubin bands | observed | chance (permutation null) |
|-------------------------------------------|----------|---------------------------|
| $k \geq 2$                                | 181      | $68 \pm 7$                |
| $k \geq 3$                                | 113      | $5 \pm 2$                 |
| $k \geq 4$                                | 61       | 0.2                       |
| $k \geq 5$                                | 23       | 0.0                       |

The null shuffles flags within (band, raw offset, S/N) classes, preserving all rates and the VIS-induced raw correlations (hence chance  $k \geq 2$  is 68, not 0). Robust to the cuts (S/N>10: 200 vs 8; all nine bands: 153 vs 12) and to the flag definition, which is a purity dial: from “above 1:1” to “2× and +15 mas” the  $k \geq 3$  excess stays 50–180 while the chance background collapses from 214 to 1: the fiducial choice gives ~95% purity.

**Stage 1 established a source-attached, band-shared cause, not yet necessarily the head.** Four things are shared; each gets a measurement:

- **(1) The VIS reference?** A bad  $\text{pos}_{\text{VIS}}$  would worsen all bands at once, a plain measurement error. Gaia says no: for the offender stars the VIS labels sit **7.0 mas** from the PM-propagated Gaia position, better than the survey average.
- **(2) The WCS?** A solution error is smooth in sky position: it cannot single out one star and spare its neighbours. Offenders show **no sky clustering** (nearest-offender distance 62'' vs  $66'' \pm 5''$  for random draws), and the failure **persists in NISP** (59% of offenders worsened in all NISP bands), whose astrometric solution is independent of Rubin's.
- **(3) The source itself?** Proper motion correlates with nothing ( $\rho = 0.01$  against residuals); high-PM stars are actually the head's success cases (it bridges the epochs).
- **(4) The head's representation** carries positive fingerprints, not just survival by elimination: corrections **larger than the disagreement they correct** ( $|c| \sim 28 \text{ mas}$  vs raw  $\sim 11 \text{ mas}$ ), directions coherent across all nine bands and **both instruments** (one wrong belief applied everywhere), and against Gaia the head sits at **22.9 mas** where every classical measurement of the same stars sits at 5–7 mas.

**The two-stage verdict:** repetition proves the cause is source-attached and shared; elimination plus fingerprints prove it is the head. The 113 are 0.3% of the catalog, identifiable one by one.



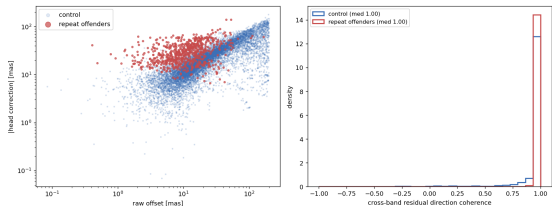
Offender rate for stars ramps 2%  $\rightarrow$  **14%** from VIS 24 to VIS 17; extended sources stay flat.

Population: the 3,185 sources with  $\geq 2$  bright (S/N>20) Rubin bands, the only ones the  $k$  statistic can score, out of 37,796 in the catalog; N labels are the per-bin denominators of that eligible population, and rate  $\times$  N summed over bins and classes recovers the 113.

Mechanisms tested and **ruled out**, each with a measurement:

- proper motion ( $\rho = 0.01$  with residuals)
- tile-edge context (paired test:  $+0.1$  mas)
- WCS patches (no sky clustering)
- companions within  $2''$  (13% vs 20% control)
- VIS saturation (onset  $\approx 17$ , masked not clipped; offender cores pristine)

Against PM-propagated Gaia DR3 (epoch 2024.83), per star:

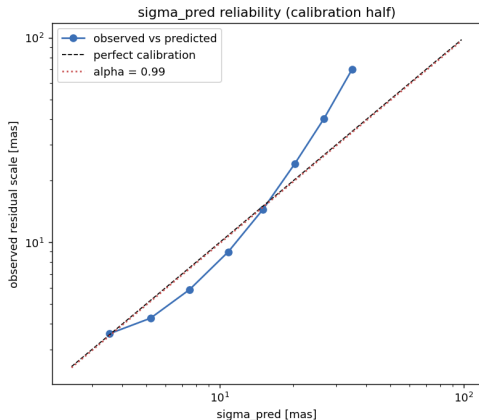


|                               | VIS label      | head     |
|-------------------------------|----------------|----------|
| offender stars ( $k \geq 3$ ) | <b>7.0 mas</b> | 22.9 mas |
| all Gaia stars                | 9.5 mas        | 12.2 mas |

Left: offenders receive corrections **larger than the raw disagreement** ( $|c| \sim 28$  mas vs raw  $\sim 11$  mas), coherently in every band: the head asserts a center the classical centroids contradict.

Reading: a brightness-dependent divergence between the Gaussian-fit convention and the head's feature-encoded center (PSF wings, training rarity,  $\sigma_{\text{label}} \equiv 5$  mas loss floor).

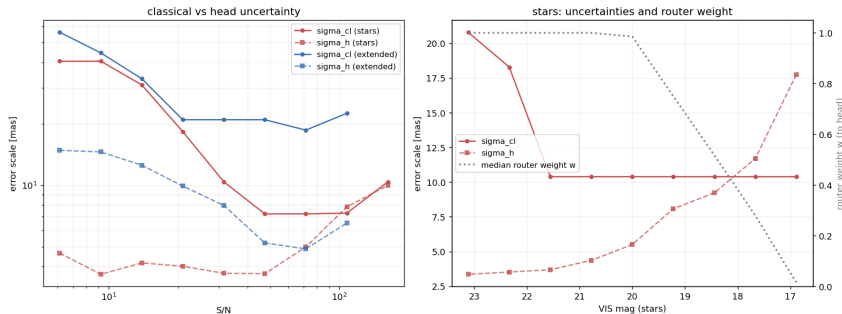
Two estimators of the same VIS-convention position: classical (error scale  $\sigma_{cl}$  from S/N and class) and head-corrected (the head's own  $\sigma_{pred}$ ).



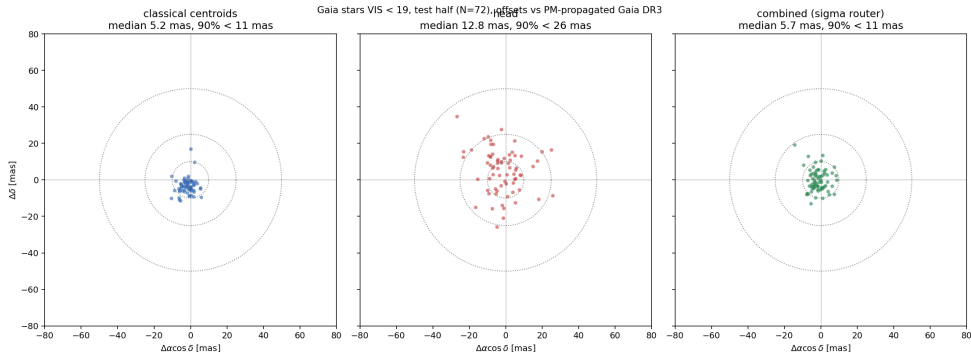
combined =  $(1 - w)$  raw +  $w$  resid,

$$w = \frac{\sigma_{cl}^2 - \rho \sigma_{cl} \sigma_h}{\sigma_{cl}^2 + \sigma_h^2 - 2\rho \sigma_{cl} \sigma_h}$$

- $\sigma_{pred}$  is nearly perfectly calibrated in bulk ( $\alpha = 0.99$ ); the over-confidence lives only in the offender tail.
- The correlation matters: head and classical errors share photons ( $\rho \approx 0.8$  extended, 0.5 stars); naive inverse variance costs 2 mas at the faint end, the correlation-aware weight costs nothing.
- All calibrations fit on half the tiles; all

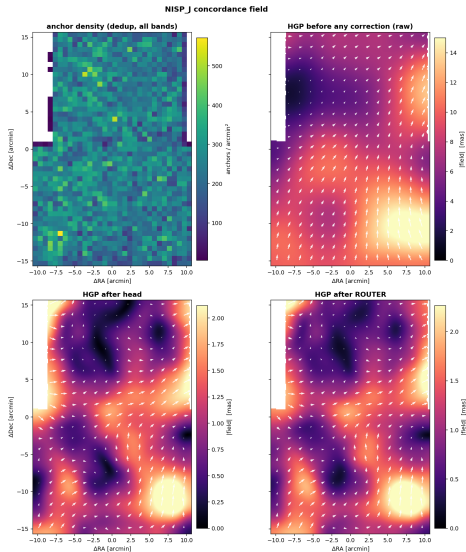


- Extended sources: classical centroiding is morphology limited ( $\sigma_{cl} \gtrsim 20$  mas at every S/N); the curves **never cross**, so galaxies are pure head.
- Stars: crossing at  $S/N \approx 108$  (VIS  $\approx 18.2$ ); the weight (gray) slides  $1 \rightarrow 0$  across VIS  $20 \rightarrow 17$ . The magnitude gate is **derived from the uncertainties**, not imposed.
- $\sigma_h$  **rises** for bright stars: the uncertainty channel partially knows the head struggles there; the offenders are the tail where that self-awareness fails.



| median radial offset [mas]  | classical  | head | router      |
|-----------------------------|------------|------|-------------|
| all sources                 | 50.6       | 14.6 | <b>14.7</b> |
| bright stars, vs Gaia truth | <b>5.2</b> | 12.8 | <b>5.4</b>  |
| faint (S/N < 10)            | 64.9       | 18.5 | <b>18.5</b> |

Bright stars made worse than classical: 38% (head) → 16% (router), with the large-amplitude



- Refit the field (HGP super-tight, PINN) on router residuals: median change **0.16 mas**, an order of magnitude below the field amplitude. The robust fit had already absorbed the 113 outliers; Figure 10 and the FITS products stand.
- The  $\sim 15$  mas lobe at lower right is **real**: coherent in 8 of 9 bands relative to VIS, normal anchor coverage, both solvers agree, and it sits within MER tile TILE102043549. Eight rulers agree and one does not: a localized feature of the **VIS solution**.
- By construction no VIS-trained correction can remove it; the released raw-field FITS is precisely the map that carries it.

## Takeaways

- Above the 1:1 line is two populations:  $\sim 77\%$  benign regression to the corrected floor,  $\sim 23\%$  real failures concentrated in bright point sources (0.3% of the catalog).
- Gaia certifies the mechanism: the head, not the labels; a brightness-dependent centroid-convention divergence on healthy stellar profiles, with PM, edges, WCS, companions, and saturation all measured and eliminated.
- The calibrated- $\sigma$  router is never worse than either estimator: head performance at the faint end, classical parity on bright stars, zero cost.
- The head improves astrometry where classical methods are noise limited and defers where they are systematics limited. That is the paper sentence.

## Next steps

- Gaia-supervised head retrain targeting parity ( $\sim 5$  mas): decides whether the router's bright-star branch is temporary or permanent.
- Residual directions vs local ePSF asymmetry: separates PSF-wings physics from training-rarity statistics.
- Confirm the TILE102043549 VIS-frame feature against a dense external Gaia sample; flag in the released field FITS.